

Ronald Saraswat, PhD

CELL & MOLECULAR BIOLOGIST

EXTRACELLULAR VESICLES • AGEING & SENESCENCE • QUANTITATIVE CELL BIOLOGY

Paris, France | [Portfolio](#) | [Google Scholar](#) | [LinkedIn](#) | [ORCID](#)



RESEARCH INTERESTS

- Ageing Biology & Cellular Senescence
- Intercellular Communication & Extracellular Vesicles
- Quantitative Proteomics & Functional Cell Biology
- Cellular Stress & Cell-state Transitions
- Endothelial & Vascular Biology

CORE EXPERTISE

- Primary human **endothelial cells** (HAoEC/HUVEC), HeLa, MSC and NSC-34 culture
- Long-term primary-cell and mammalian-cell models, transfection and functional assays
- **Flow cytometry**, confocal microscopy and quantitative imaging
- Immunofluorescence
- Cellular stress, **senescence** and cell-cycle readouts
- EV isolation: differential centrifugation and SEC
- NTA, TRPS/qNano, zeta potential and DLS
- S-Trap sample preparation and **LC-MS/MS proteomics**
- MaxQuant, Perseus, **R/Bioconductor**, Cytoscape, Reactome and GraphPad Prism

ANIMAL RESEARCH

CESU Level B (France) and LAST-certified in Ireland for rodent experimentation. Practical training includes intravital EV imaging and vena cava ligation; *theoretical and husbandry training in aquatic models*.

AWARDS & LEADERSHIP

- **Young Investigator Award**, TiCardio Training School, Maastricht (2022)
- Marie Skłodowska-Curie Actions Doctoral Fellow, TiCardio ITN (2020–2023)
- **Elected Early-Stage Researcher Representative**, TiCardio Supervisory Board (2021–2023)
- Co-founder and organiser, Gene & Tonic Cité (2026–present)

EDUCATION

PhD, Health Biology

Aix-Marseille Université & Johannes Gutenberg University Mainz | Dec 2023

MSc, Neuropharmacology (2.1)

University of Galway | 2017

BSc, Life Sciences

University of Delhi | 2015

University Diploma, Student Entrepreneurship (D2E1)

Aix-Marseille Université | 2024

LANGUAGES

English: Native / Fluent

French: A2-B1 (Intermediate)

SCIENTIFIC PROFILE

MSCA-trained cell and molecular biologist with a PhD in Health Biology and postdoctoral experience in mammalian cell biology, extracellular vesicles, quantitative functional assays and LC-MS/MS proteomics. Experienced in combining primary human cell models, flow cytometry, confocal microscopy and quantitative proteomics to investigate mechanisms of ageing, cellular senescence and intercellular communication. My research interests centre on cellular ageing and stress, extracellular-vesicle-mediated communication, quantitative cell biology and the translation of molecular findings into measurable cellular phenotypes.

RESEARCH EXPERIENCE

Postdoctoral Researcher | Apr 2025 - Mar 2026

Lavie Lab, Université Paris Cité | Paris, France

- Optimised and standardised quantitative NanoLuc-based workflows for reproducible assessment of extracellular-vesicle-mediated cargo delivery and intracellular functional readouts.
- Optimised plasmid preparation, Lipofectamine transfection, EV dosage and assay conditions in donor–recipient systems using artificial-fusogen overexpression.
- Evaluated uptake and functional intracellular cargo delivery of modified and lipid-hybridised EVs in a cross-institute translational collaboration.
- Coordinated two parallel projects, including experimental planning, structured troubleshooting, documentation, result review, scientific communication and guidance of Master's and PhD researchers.

Doctoral Researcher - MSCA TiCardio ITN | Jan 2020 - Dec 2023

Aix-Marseille Université, C2VN & Johannes Gutenberg University Mainz, CTH | France / Germany

- Investigated how ageing and replicative senescence alter primary human endothelial cells, endothelial EVs and communication with recipient cells.
- Established long-term HAoEC and HUVEC models and quantified cellular stress, senescence and endothelial activation using SA-beta-galactosidase, gamma-H2AX, cell-cycle/stress markers, flow cytometry, immunofluorescence and confocal microscopy.
- Isolated and characterised EVs by differential centrifugation, size-exclusion chromatography, NTA, TRPS/qNano, zeta potential/DLS, microscopy and flow-based approaches.
- Assessed thrombo-inflammatory mechanisms using tissue-factor activity, thrombin-generation assays, TNF-alpha priming and recipient-cell functional experiments.
- Prepared S-Trap samples for quantitative LC-MS/MS proteomics and analysed datasets using MaxQuant, Perseus, R, Cytoscape, Reactome and FunRich.
- Identified age-associated extracellular-vesicle signatures, prioritised candidate proteins for mechanistic validation and contributed to a first-author manuscript currently under peer review.

Earlier Research Experience

- Research Assistant, OmniSpirant/University of Galway (2019): engineered MSCs, EV therapeutics, lentiviral GFP transduction, NTA, DLS and PEGylation.
- Research Assistant, University of Galway (2017–2018): redox signalling, mitochondrial biology and quantitative protein analysis.

SELECTED PUBLICATIONS AND RESEARCH OUTPUTS

- Saraswat R. et al. First-author original research manuscript on age-associated endothelial extracellular-vesicle proteomics and function. Under peer review (2026). *Contact via website contact form for more details.
- Akbulut A.C., Saraswat R. et al. “*Blood Coagulation and Beyond.*” *Thrombosis and Haemostasis*. 2023;123(8):808-839. DOI: 10.1055/a-2052-9175.
- Konkoth A., Saraswat R. et al. “*Multifaceted Role of Extracellular Vesicles in Atherosclerosis.*” *Atherosclerosis*. 2021;319:121-131. DOI: 10.1016/j.atherosclerosis.2020.11.006.
- Saraswat R., McDonagh B. “*Diagonal Electrophoresis for the Detection of Proteins Involved in Disulfide Bonds.*” *Methods Mol Biol*. 2019;1855:279-286. DOI: 10.1007/978-1-4939-8793-1_23.

PRESENTATIONS, LEADERSHIP & PROFESSIONAL ACTIVITIES

- **Oral presentation:** Senescence Molecular Signature on Endothelial Extracellular Vesicles, Joint TiCardio–INNOVTE Conference, Marseille, France (2023).
- **Poster presentation:** Role of Endothelial Extracellular Vesicles in the Propagation of Pro-Senescence Signals and Thrombo-Inflammation, TiCardio Training School, Maastricht, The Netherlands (2022) – **Young Investigator Award**.
- **Co-founder and organiser:** Gene & Tonic, Cité Internationale Universitaire de Paris (2026–present).

RESEARCH MOBILITY AND COLLABORATION

- MSCA doctoral secondments at CTH Mainz (cellular senescence and vascular thrombosis models and assays, intravital EV imaging, and mouse vena cava ligation training) and ISAS Dortmund (quantitative LC-MS/MS proteomics).
- Cross-laboratory project coordination and scientific collaboration across France, Germany and Ireland.